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BYD All-Solid-State Battery Commercialization Timeline

What public primary sources actually support about the reported 2027 demonstration and around-2030 mass-production milestones

PUBLIC RESEARCH REPORT

Information cut-off **17 July 2026**

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Scope Publicly accessible corporate filings, official event reporting, standards records and patent publications

This report is informational and intended for pre-diligence. It is not investment, legal, technical-certification or procurement advice.

01 The short answer

Bottom line

Public evidence confirms a long-running and technically broad BYD research program in solid electrolytes and all-solid-state cell, pack and vehicle-integration concepts. An official conference report also records a BYD battery executive stating a plan to begin batch demonstration vehicle deployment around 2027 and to pursue large-scale production around 2030. However, the public primary-source record does not disclose the pilot-line capacity, process yield, independently verified cell performance, named vehicle program, supplier qualification, unit cost or capital commitment needed to underwrite either milestone as an execution-ready production schedule.

Assessment of the 2027 milestone: Plausible, but not verifiable from the current public primary-source record.

Confidence: low to medium. The target has an attributable executive source and BYD has relevant patents, but the execution bridge from invention to automotive demonstration is not publicly evidenced.

Assessment of the around-2030 milestone: A credible strategic direction, but not a forecast-ready scale-production date. Confidence: low. No public cost model, qualified supply chain, industrial-yield evidence, vehicle validation record or production-capacity disclosure was located.

Most important analytical distinction: patents establish technical activity and problem-solving intent; they do not establish technology readiness level, cell performance, pilot output, production yield, cost, homologation or launch timing.

Decision dashboard

Long-running R&D CONFIRMED Patent priorities date to 2015 and cover electrolyte, electrode, cell, pack and vehicle-pressure topics.	Reported 2027 target ATTRIBUTABLE An official conference report attributes the target to BYD CTO Sun Huajun; no BYD filing or official transcript was located.	Pilot line / yield NOT DISCLOSED No primary public source was found for GWh capacity, commissioning status, process yield or throughput.
Cell performance NOT VERIFIED No public test certificate or report was found for energy density, cycle life, safety or temperature performance.	Named vehicle program NOT DISCLOSED No primary source was found naming a model, fleet size, delivery quarter or homologation program.	Around-2030 scale NOT FORECAST-READY The direction is reported; the manufacturing, supply and cost evidence required for scale is absent.

What this report does - and does not do

- Uses a primary-source-first hierarchy: corporate filings, official conference reporting, national standards records, patent publications and peer-company official disclosures.
- Separates an executive target from a formal corporate production commitment.
- Treats absence from statutory filings as an absence of public disclosure, not proof that an internal program does not exist.
- Uses selected patents to establish technical breadth; it is not a complete freedom-to-operate or patent-landscape analysis.
- Does not use repeated media articles, social posts or anonymous claims as independent evidence.

02 Research question and evidence rules

Research question

Question

What do publicly accessible primary and primary-adjacent sources support about BYD's reported plan to begin batch demonstration vehicle deployment of all-solid-state batteries around 2027 and achieve large-scale production around 2030?

Evidence classification

Classification	Meaning
Confirmed	A directly accessible corporate, regulatory, patent or government source establishes the fact.
Attributable executive statement	An official event report or transcript attributes the statement to a named company executive; it is not the same as a board-approved filing or product commitment.
Analytical inference	A reasoned interpretation from multiple confirmed facts. The underlying sources do not state the conclusion directly.
Not publicly disclosed	No qualifying primary source was located within the defined search scope. This does not prove the information does not exist internally.
Unsupported claim	A claim found in secondary circulation but not supported by a qualifying primary source and therefore excluded from the assessment.

Source review performed

- BYD statutory disclosures: 2025 interim report, 2025 annual report and 2026 first-quarter report. Each was text-searched in Chinese and English for solid-state battery terminology.
- Official conference reporting for the February 2025 China all-solid-state battery collaborative innovation platform annual meeting.
- National standards project records for EV solid-state battery terminology, performance and life specifications.
- Selected BYD patent publications with priorities from 2015 through 2024, including materials, electrode, cell/pack and vehicle pressure-management topics.
- Official Toyota, Nissan and Samsung SDI disclosures used only as peer-evidence calibration - not as proof of BYD's progress.

Important limitations

- No official BYD presentation deck, full video or verbatim transcript from Sun Huajun's February 2025 speech was located. The official event report is therefore the best attributable public record found for the milestone wording.
- Patent databases describe inventions and publication metadata; they do not demonstrate that the claimed embodiments were built at automotive scale or selected for production.
- Public-company reports may omit immaterial or pre-commercial programs. Non-disclosure must not be interpreted as non-existence.
- No confidential supplier, customer, laboratory or internal program data was used.

03 What is actually reported about 2027 and 2030

Attributable source

An official event report published by Southeast University and sourced to Xinhua records BYD Lithium Battery CTO Sun Huajun speaking at the February 2025 annual meeting of China's all-solid-state battery collaborative innovation platform. The report says BYD had begun feasibility validation covering key materials, cell-system development and production-line construction, planned batch demonstration vehicle deployment around 2027, and expected large-scale production around 2030. [1]

Interpretation discipline

The source supports an attributable executive target. It does not disclose a quarter, fleet size, vehicle model, factory location, line capacity, cell specification, production yield or customer delivery commitment. Those more precise claims should not be treated as confirmed unless an additional primary source is produced.

Milestone / claim	Public-source wording	Evidence type	Current use	Confidence
Around 2027	"Batch demonstration installation" (official event report paraphrase/quotation)	Named executive statement	Directional program target	Low-Medium
Around 2030	"Large-scale production" (official event report paraphrase/quotation)	Named executive statement	Long-term scale ambition	Low
2027 Q1	No qualifying primary source located	Not publicly disclosed	Do not use	Not rated
1,000 vehicles / named models	No qualifying primary source located	Not publicly disclosed	Do not use	Not rated
Specific GWh lines / factories	No qualifying primary source located	Not publicly disclosed	Do not use	Not rated

Corporate filing cross-check

The 2025 interim report, 2025 annual report and 2026 first-quarter report were searched for "固态电池", "全固态", "solid-state" and close variants. No matches were found. [5-7] This indicates that the program and milestones were not specifically described in those statutory reports; it does not establish that BYD has no internal program.

For diligence purposes, the distinction matters: a conference target can be useful as management direction, but it carries less commitment and detail than a disclosed production plan, named capital project, customer program or board-level filing.

04 What BYD's patent record confirms

Selected patent publications confirm that BYD has worked on solid electrolyte and all-solid-state battery problems for at least a decade. The technical scope appears to have broadened from electrolyte materials and interfaces toward complete cell/pack architecture and vehicle pressure management. This is meaningful evidence of sustained R&D, but it is not a production-readiness certificate.

Priority date	Publication	Subject	What it directly shows	Read-through
2015-06-25	WO2016206430A1	Solid electrolyte and lithium-ion battery	Oxide/NASICON-type core-shell electrolyte and electrochemical-window problem	Materials research
2016-05-31	CN107452983B	Composite electrolyte and lithium-ion battery	Sulfide solid electrolyte combined with plastically deformable organic lithium salt to address grain-boundary/interface resistance	Sulfide-interface research
2016-07-29	WO2018019238A1	Negative electrode material and all-solid-state lithium-ion battery	Negative-electrode architecture for all-solid-state cells	Electrode/cell research
2020-07-13	CN113937346B	Solid electrolyte and all-solid-state battery	Mixed oxide-sulfide glassy system targeting conductivity, stability and strength	Hybrid electrolyte research
2024-08-07	CN118748295B	All-solid-state battery, battery pack and electrical equipment	Cell outer-layer/ceramic architecture for pressure handling and cycle performance	Cell/pack architecture
2024-08-05	WO2026031516A1	Pressure regulation system, pressure-management method and vehicle	Vehicle-level pressure-management concept for solid-state batteries	Vehicle integration concept

Reasonable inference from the patent sequence

- High-confidence inference: BYD has a persistent solid-state research capability spanning more than one technical route and more than one layer of the battery system.
- Medium-confidence inference: newer filings indicate interest in practical cell construction and vehicle integration, not only material discovery.
- Not supported: that any patent is used in a production-intent cell, that a pilot line is operating, or that a 2027 vehicle program has passed qualification.

Patent evidence is necessary but insufficient

A patent can be filed before, during or after laboratory work and may never be commercialized. It does not disclose stable production yield, production-capable equipment, validated automotive life, qualified suppliers, cost per kWh or a launch decision.

05 What the public record does not yet establish

Standards status is still evolving

China's national project for "Solid-state battery for electric vehicle - Part 1: Terms and classification" is listed as a recommended national-standard project, not a published mandatory standard, with a proposed implementation date of 1 January 2028. Related performance and life specifications are also listed as projects. [2-4] Therefore, claims that a BYD cell has already received certification under a completed national all-solid-state battery framework require a separate, directly accessible certificate and test standard.

Readiness dimension	Public evidence located	Assessment	Critical limitation
Program direction	Attributable executive target plus sustained patents	Partially evidenced	Executive target is not a production commitment
Electrolyte/material R&D	Multiple BYD patent publications	Evidenced	Patents do not disclose automotive-scale reproducibility
Cell architecture	Recent all-solid-state cell/pack patent	Evidenced concept	No disclosed production-intent specification
Vehicle integration	Pressure-management vehicle patent	Evidenced concept	No named model or validation program
Independent performance test	No qualifying report/certificate located	Not disclosed	Energy density, cycle life, safety and low-temperature claims remain unverified
Pilot line and capacity	No qualifying corporate/government source located	Not disclosed	No GWh capacity, site, commissioning or output data
Manufacturing yield	No qualifying source located	Not disclosed	No continuous-run yield, throughput or equipment effectiveness
Supplier qualification	No qualifying source located	Not disclosed	No named production-intent materials/equipment supply chain
Vehicle validation	No qualifying source located	Not disclosed	No test mileage, durability, homologation or fleet data
Cost and scale economics	No primary cost model located	Not disclosed	No current cost/kWh or path to parity

Claims excluded from this assessment

The following precise claims were not used because no qualifying primary source was located during this review: a Shenzhen 2 GWh pilot line; Chongqing 20 GWh or 80 GWh all-solid-state lines; 95% or 99.2% yield; 1 million kilometres of testing; a 1,000-vehicle 2027 fleet; Yangwang U9, Han EV or Denza D9 deployment; 10,000-cycle life; 32°C nail-penetration temperature; 12-item automotive certification; 70% equipment compatibility; or a specific cost-parity figure.

These claims may be investigated in a future expanded project, but they should remain "unverified" until supported by original corporate, government, testing, tender, environmental-review or supplier-disclosure records.

06 2027 batch demonstration: plausible, not underwritten

Evidence in favour

- A named BYD battery executive is officially reported as setting an around-2027 batch demonstration objective. [1]
- BYD's patent record confirms longstanding activity in electrolyte, interface, cell and integration problems relevant to all-solid-state batteries. [8-13]
- Recent patent priorities in 2024 indicate continued work at cell/pack and vehicle-system levels rather than a dormant historical portfolio. [12-13]
- An around-2027 demonstration objective is not inherently outside the global industry range: Toyota and Samsung SDI publicly target 2027-28 commercialization/mass production, while Nissan targets an FY2028 vehicle and publicly reports a pilot line. [14-18]

Evidence missing

- A disclosed BYD pilot line, capacity, commissioning date and stable output.
- A production-intent cell specification and independent test report tied to a traceable sample.
- A named vehicle program, fleet quantity, validation plan or homologation status.
- Automotive qualification data: cycle/calendar life, fast-charge durability, safety, pressure control and temperature performance under defined protocols.
- Qualified material and equipment suppliers, or evidence of internal industrial capacity.
- Manufacturing yield, process capability and a signed product or capital commitment.

2027 assessment

Low-Medium confidence. A limited demonstration around 2027 is plausible as a program objective, but the present public evidence cannot establish that BYD is on schedule, define the scale, or identify a production vehicle. The threshold for upgrading the assessment is direct execution evidence - not additional media repetition of the same target.

07 Around-2030 large-scale production: direction credible, date not supported

The official event report establishes an around-2030 large-scale-production ambition. [1] The patent record establishes relevant R&D breadth. Those facts support monitoring BYD as a serious participant. They do not support a forecast of commercial scale by a specific date.

Why the confidence is lower than for 2027

- Scale requires stable industrial yield and throughput, not only working cells.
- Sulfide or mixed solid-electrolyte supply must be produced at automotive quality, cost and volume.
- Vehicle integration must work across pressure, thermal, safety, serviceability and pack-manufacturing requirements.
- Cost parity requires a transparent materials, equipment, yield and depreciation model. No BYD primary-source model was located.
- A demonstration fleet can be management-controlled; broad commercialization depends on suppliers, regulation, customer acceptance and product economics.

Around-2030 assessment

Low confidence as a dated scale outcome. The ambition is attributable, but the current primary-source record is not sufficient to forecast volume, cost parity, model coverage or market share. The milestone should be treated as a strategic monitoring horizon rather than a base-case production assumption.

08 What stronger public execution evidence looks like

Peer disclosures are not used to rank BYD. They are used to calibrate the type of evidence normally available when a company moves from research intent toward commercialization.

Company	Public target	Public execution evidence	Calibration read-through
Toyota / Idemitsu	2027-28 market introduction / commercialization; later full-scale production	Official joint agreement; sulfide focus; phases for pilot facility, mass-production technology and supply chain; METI-certified development/production plan	Shows named partner, phased industrial plan and government certification.
Nissan	FY2028 vehicle target	Officially displayed pilot line; states operation began January 2025; production-process partnership for dry electrodes	Shows physical pilot-line status and a process-development workstream.
Samsung SDI	Mass production targeted for 2027 / second half 2027	Official 6,500 m ² S-Line; samples supplied to customers; 900 Wh/L stated; validation collaboration with BMW/Solid Power	Shows pilot facility, samples, customer testing and commercialization team activity.
BYD	Around-2027 batch demonstration; around-2030 large-scale production	Official event report plus patents; no public pilot-line, customer sample, vehicle program or test certificate located	Shows credible intent and R&D, but a thinner execution-evidence layer.

This comparison does not prove that peers are technically ahead. It shows that public evidence quality differs. BYD may possess undisclosed capabilities; a public-source diligence assessment cannot credit them until evidence becomes accessible.

09 What would change the assessment

Priority	Trigger	Evidence required
1	Pilot line	Official disclosure of location, designed capacity, commissioning status and production-intent purpose.
2	Traceable cell specification	Cell capacity, gravimetric/volumetric energy density, chemistry, pressure condition, test method and sample identity.
3	Independent validation	A complete report or certificate from a named laboratory under an identified standard and sample configuration.
4	Manufacturing evidence	Continuous-run output, yield, process capability, equipment readiness and batch consistency.
5	Vehicle program	Named platform/model, fleet size, integration status, test mileage and homologation pathway.
6	Supply chain	Named qualified suppliers or disclosed internal capacity for electrolyte, cathode/anode and production equipment.
7	Economics	A cost bridge covering materials, yield, capex/depreciation, pressure-management hardware, pack design and recycling.
8	Recurring corporate disclosure	Inclusion in annual/interim reports, investor materials, government project approvals or formal capital plans.

Practical monitoring rule

Upgrade the 2027 assessment only when at least two independent execution layers appear - for example, a disclosed pilot line plus a traceable automotive test report, or a named vehicle program plus qualified supplier/capacity evidence. A new patent or another repetition of the same target is not enough.

DECISION IMPLICATIONS

10 Conclusions for pre-diligence

Question	Answer	Reason
Is BYD seriously researching all-solid-state batteries?	Yes - high confidence.	A decade-long selected patent record spans materials, interfaces, cell/pack architecture and vehicle pressure management.
Did a named BYD executive communicate a 2027/2030 direction?	Yes - medium-high confidence.	An official conference report attributes the targets to BYD Lithium Battery CTO Sun Huajun.
Is a 2027 Q1 / 1,000-vehicle / named-model plan confirmed?	No.	No qualifying primary source was located for these precise claims.
Can current evidence establish 2027 readiness?	No.	Pilot, yield, cell validation, vehicle and supplier evidence are not publicly disclosed.
Can current evidence support around-2030 scale production as a base case?	No.	Cost, industrial yield, supply scale and vehicle-program evidence are missing.
Should BYD be monitored?	Yes.	The combination of executive intent and recent technical filings justifies continued monitoring.

Recommended use of this assessment

- Investment research: treat the 2027/2030 milestones as management direction, not forecast-ready production commitments.
- Corporate strategy: monitor BYD's public execution signals and compare evidence depth, not only announced dates.
- Supplier strategy: do not assume a production-intent supplier relationship without a filing, contract, project approval or attributable company disclosure.
- Media/editorial use: distinguish patents, executive statements, demonstrated cells, pilot production, vehicle validation and mass production as separate maturity stages.

Final assessment

VoltChina view

BYD has a credible all-solid-state battery research program and an attributable commercialization ambition. Public evidence is strong enough to justify monitoring and deeper diligence, but not strong enough to validate a specific 2027 execution plan or an around-2030 scale-production forecast. The central diligence question is no longer whether BYD has relevant inventions; it is whether BYD can publicly demonstrate reproducible automotive cells, industrial process control and a qualified path from cell to vehicle.

11 Primary and official sources reviewed

Titles below are hyperlinked. Access dates: 17 July 2026. Patent legal-status fields shown by aggregation services should be independently confirmed in the relevant patent office for legal reliance.

Ref	Date / priority	Source	Document	Use in report
1	2025-02-23	Southeast University / Xinhua	Official event report: 2025 China all-solid-state battery collaborative innovation platform annual meeting	Milestone attribution and program scope
2	Current record	SAMR / SAC	National standard project: EV solid-state battery Part 1 - terms and classification	Status and proposed implementation date
3	Current record	SAMR / SAC	National standard project: EV solid-state battery Part 2 - performance specification	Standards pipeline
4	Current record	SAMR / SAC	National standard project: EV solid-state battery Part 4 - life specification	Standards pipeline
5	2026-03-27	BYD / HKEX	BYD Company Limited 2025 Annual Report	Corporate disclosure cross-check
6	2025-08-30	BYD / CNINFO	BYD 2025 Interim Report	Corporate disclosure cross-check
7	2026-04-29	BYD / CNINFO	BYD 2026 First-Quarter Report	Corporate disclosure cross-check
8	Priority 2015-06-25	BYD patent publication	WO2016206430A1 - Solid electrolyte and preparation method; lithium-ion battery	Early solid-electrolyte research
9	Priority 2016-05-31	BYD patent publication	CN107452983B - Composite electrolyte and lithium-ion battery	Sulfide electrolyte/interface research
10	Priority 2016-07-29	BYD patent publication	WO2018019238A1 - Negative electrode material and all-solid-state lithium-ion battery	Electrode/cell research
11	Priority 2020-07-13	BYD patent publication	CN113937346B - Solid electrolyte and all-solid-state battery	Mixed oxide-sulfide electrolyte research
12	Priority 2024-08-07	BYD patent publication	CN118748295B - All-solid-state battery, battery pack and electrical equipment	Cell/pack architecture

Ref	Date / priority	Source	Document	Use in report
13	Priority 2024-08-05	BYD patent publication	WO2026031516A1 - Pressure regulation system for solid-state battery and vehicle	Vehicle integration concept
14	2023-10-12	Toyota / Idemitsu	Cooperation toward mass production of all-solid-state batteries for BEVs	Peer evidence calibration
15	2024-09-06	Toyota	Next-generation battery development and production plan certified by METI	Peer evidence calibration
16	2025-08-19	Nissan	LiCAP partnership and all-solid-state battery pilot-line status	Peer evidence calibration
17	2024-12-03	Samsung SDI	900 Wh/L all-solid-state battery and S-Line sample status	Peer evidence calibration
18	2025-10-31	Samsung SDI	All-solid-state battery validation project with BMW and Solid Power	Peer evidence calibration

Search terms and negative-source checks

Corporate-report searches included: 固态电池, 全固态, 固态, solid-state and solid state. No solid-state-battery references were found in the reviewed 2025 interim report, 2025 annual report or 2026 first-quarter report. The reports do contain broad statements on battery R&D and technology investment, which were not treated as solid-state-specific evidence.

The review also attempted to locate public primary evidence for specific pilot-line, yield, vehicle, test and cost claims. Where no qualifying source was located, the report records “not publicly disclosed” rather than assuming the claim is false.

Disclaimer

This report is based on sources accessible to the public as of the information cut-off. It is a pre-diligence assessment, not a complete technical, legal, patent, accounting or investment due-diligence opinion. Translations and interpretations should be checked against the original-language source for transaction or litigation use. Future disclosures may materially change the assessment.

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